OPERATIONAL RISK ANALYSIS:

PORTFOLIO PROJECT: SETTLEMENT QUERY MANAGEMENT DASHBOARD

Executive Summary:

This project involves the creation of an interactive dashboard and Aged Risk Tracker for a Markets Operations environment. The tool is engineered to provide real-time visibility into trade settlement queries, prioritising "High Risk" to ensure regulatory compliance. By transitioning from static data lists to a dynamic, risk-first visualization, the dashboard reduces management blind spots and accelerates decision-making.

Problem Statement:

In high-volume financial operations, client queries can be spread across multiple desks. Manual tracking leads to "Aged Risks" - trades that remain unsettled past their value date, increasing financial exposure.

Objectives:

- **Centralisation** - Consolidate trade queries into a single resource.
- **Mitigate Risk** - Implement logic to categorise trades into Low, Medium and High Risk buckets and flag High Risk buckets.
- **Dynamic Interactivity** - Enable desk-specific drill downs using synchronised slicers.

Technical Architecture

The Settlement Query Manager was built with 3 key architecture principles in mind: Scalability, Data Integrity and User experience.

Dynamic Scalability

The raw data was converted into an Excel Table, the use of a Table ensures that any new queries added automatically inherit all the formulas and pivot tables, thereby eliminating the need for manual adjustments.

Data Integrity

The NETWORKDAYS function is used to filter out weekends and holidays, ensuring the dashboard reflects market days rather than calendar days.

Aged days are calculated using a nested IF function which states that for active items, calculate the age against the TODAY() function. For resolved items the age is anchored to the fixed resolution date.

A -1 offset ensures elapsed time accuracy, while a MAX(0,) wrapper stops technical errors from weekend dates.

User Experience

Pivot Tables are used to aggregate data by Asset Class, Counterparty and Risk Bucket. These are connected to slicers allowing for synchronisation and firm wide filtering.

To prevent UI errors such as when a status type has zero values, a logic bridge was created using GETPIVOTDATA and IFERROR. The formula catches the error and the dashboard displays “0”, ensuring the UI remains stable.

Filters have been applied to exclude “Resolved” data, to ensure the dashboard is focused on active financial exposure.

Data Analysis & Business Impact:

High Risk Aging

20% of total outstanding queries (10 out of 50) have moved into the High-Risk Bucket (4+ business days without being resolved).

These queries represent the highest probability of trade failures or potentially leading to regulatory fines. Queries such as CITI – 040 and CITI – 042, have reached a critical age of 5 days, requiring immediate senior escalation to prevent financial loss or reputational damage.

Counterparty Performance

Morgan Stanley and J.P Morgan contain the most High-Risk queries (2). While Morgan Stanley represents high volume, it also shows a trend of queries sitting in "Open" or "Pending" status for 4 or more days.

The dashboard highlights a need for a closer inspection as to the reason why these delays are frequently happening. For example, there may be a communication gap in the trade confirmation process.

Risk by Asset Class

The Equities and Derivative asset classes contain the highest number of aged items (3 each), while both queries in the Securities asset class have fallen into the High-Risk bucket.

The standard time for a query to settle has been set at less than 4 days. With these asset classes having many High-Risk queries and especially with all of Securities queries being High-Risk, this suggests that resources should be allocated to these desks to clear the backlog.

Performance Highlights

The use of the MAX(0,) logic successfully filtered weekend errors, ensuring Data Integrity and that the Risk Buckets are not inflated.

The teams successfully resolved 6 queries on the same day they were received, proving that the internal workflow can be highly efficient.

Troubleshooting & Root Cause Analysis:

This section includes the technical issues that I encountered during the development of the Settlement Query Manager and the logic I used to resolve them.

UI Stability – KPI cards displaying incorrect values

The issue: The KPI cards displaying the operational status (open, pending, closed) and High-Risk queries, originally displayed incorrect or a blank value, when a slicer filter was selected.

Root Cause: This occurred when the connected Pivot Table was filtered, for example, for High-Risk items, but zero trades met that criteria. In Excel when a filtered category has no data, the Pivot Table then shrinks, causing any direct cell references to those results to break.

The Resolution: To resolve this, I engineered a Logic Bridge on a secondary Calculation Sheet. Instead of linking the card displaying the KPI, directly to the Pivot Table, I used a combination of IFERROR and GETPIVOTDATA. This means that if the High-Risk data disappears, the formula catches the error and the dashboard displays "0". This ensures that the UI remains stable and reliable, even during periods of perfect operational performance.

Aging Logic – Inflated Aged Risk metrics

The issue: When I initially tested the NETWORKDAYS function, it resulted in inflated aging metrics. For example, a query that was received on Monday and resolved on Tuesday was returned as 2 days. This was a big problem as it pushed "Low Risk" into "Medium Risk" categorising queries into an incorrect Risk Bucket.

Root Cause: After doing some research I found out that Excel's NETWORKDAYS function is inclusive, counting both the start and end dates as full working days.

The Resolution: To fix this I implemented a -1 offset logic, to change the metric being measured from "Calander counting" to "Elapsed time". This means that same day resolutions are show as 0 and overnight resolutions are shown as 1, accurately reflecting true operational lag.

Additional Step: The only problem with this is if a query was received on a weekend it would cause a negative value (-1), To resolve this I wrapped the formula in a MAX(0,) function, ensuring data integrity across all 7 days of the week.

By carrying out these fixes, the project was transformed from a basic spreadsheet into a resilient business tool. The dashboard provides 100% accurate risk metrics and a dynamic UI that doesn't require manual maintenance when data volume changes.

Key Learning:

Data Engineering and Market Logic

Before this project I primarily used standard date subtraction for time-based tracking (Resolution Date – Date Received). Building this project introduced me to the importance of Market Aligned Logic, using the NETWORKDAYS function.

My biggest learning curve was resolving the counting discrepancy, realising that Excel's default counting included both the start and end dates as full working days. This lead to inflated risk metrics, such as turning a routine overnight resolution into a two-day delay.

To fix this I researched and implemented a -1 offset logic, turning the dashboard from calendar counting to an elapsed time model. To ensure trade queries received over the weekend didn't generate negative values, I learnt to account for "edge cases" applying a MAX(0,) wrapper to the formula.

This process taught me that data analysis isn't about applying formulas, it's about ensuring the underlying logic reflects the real-world operational cycles of financial markets.

Conclusion:

The development of the Settlement Query Manager successfully moves the trade exception handling process from a dispersed, reactive process into a centralised, data driven dashboard. Through process engineering, the dashboard provides a view of the firm's overall financial exposure.

The technical challenges overcome during this project included a logical bridge for UI Stability and the MAX(0,) function for correctly counting Business Days, ensuring data integrity even as volume increases and market conditions fluctuate.

Ultimately the project highlights the impact of Aged Risk Management in a Markets Operations environment. Through isolating High-Risk queries, with individual Asset classes and Counterparties, this tool helps management to move resources proactively. The move to change from managing data to managing risk is essential to reduce trade fails, avoid regulatory penalties and maintain the operational excellence required at a global institution like Citi.